

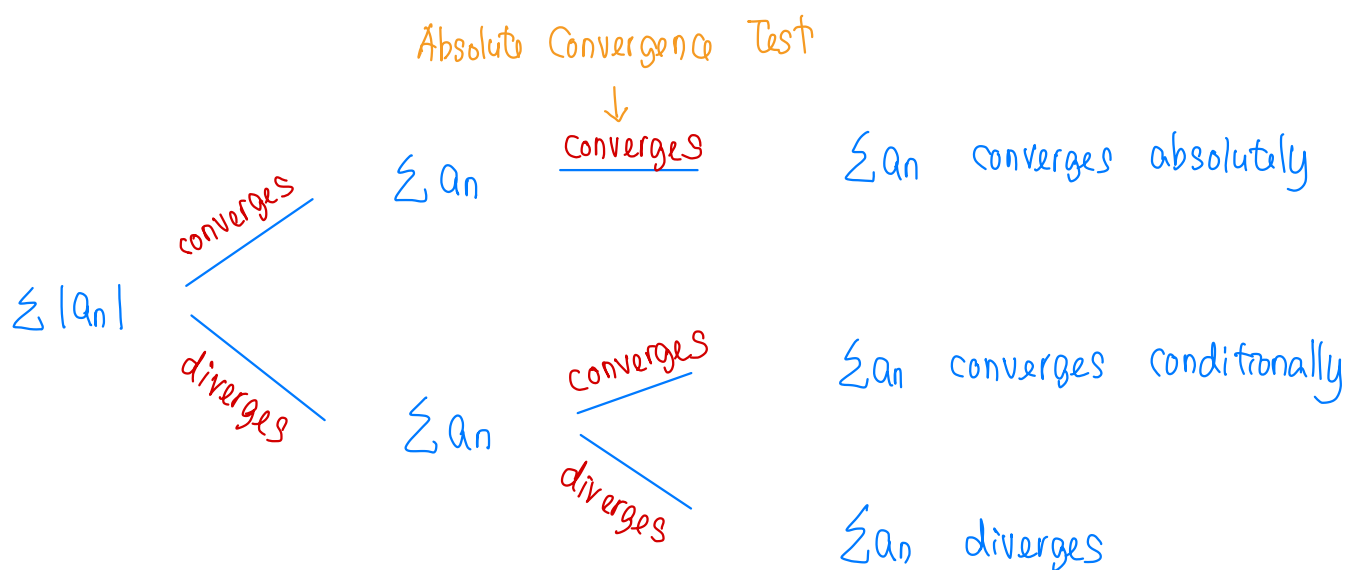
11.6 Absolute Convergence and the Ratio and Root Tests

• Definition

- If $\sum a_n$ and $\sum |a_n|$ both converge, we say that $\sum a_n$ is **absolutely convergent**.
- If $\sum a_n$ converges but $\sum |a_n|$ diverges, we say that $\sum a_n$ is **conditionally convergent**.

• Absolute Convergence Test

If $\sum |a_n|$ converges, then $\sum a_n$ converges.



- **Example** Show that $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^2}$ converges.

$$\sum_{n=1}^{\infty} \left| (-1)^{n-1} \frac{1}{n^2} \right| = \sum_{n=1}^{\infty} \frac{1}{n^2} \quad \text{converges as it is a p-series with } p=2 > 1.$$

By Absolute Convergence Test, $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^2}$ converges absolutely.

- **Example** Determine whether the series $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ is convergent.

$$\left| \frac{\cos n}{n^2} \right| \leq \frac{1}{n^2}$$

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent as it is a p-series with $p=2 > 1$.

By Ordinary Convergence Test, $\sum_{n=1}^{\infty} \left| \frac{\cos n}{n^2} \right|$ is also convergent.

By Absolute Convergence Test, $\sum_{n=1}^{\infty} \frac{\cos n}{n}$ converges absolutely.

- **Example** Determine whether the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt{n}}$ is absolutely convergent, conditionally convergent, or divergent.

$$\sum_{n=1}^{\infty} \left| (-1)^{n+1} \frac{1}{\sqrt{n}} \right| = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

This is a p -series with $p = \frac{1}{2} \not> 1$, so the series diverges.

Use Alternating Series Test :

$$\text{let } b_n = \frac{1}{\sqrt{n}}$$

$$\bullet \quad \frac{1}{\sqrt{n}} > \frac{1}{\sqrt{n+1}} \quad \text{for } n \geq 1$$

$$\bullet \quad \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$$

$$\therefore \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt{n}} \text{ converges conditionally.}$$

- **Absolute Ratio Test**

Let $\sum a_n$ be a series of nonzero terms and suppose that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$$

- (i) If $L < 1$, the series converges absolutely.
- (ii) If $L > 1$ (including $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \infty$), the series diverges.
- (iii) If $L = 1$, the test is inconclusive.

- **Example** Show that $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3^n}{n!}$ converges absolutely.

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+2} \frac{3^{n+1}}{(n+1)!}}{(-1)^{n+1} \frac{3^n}{n!}} \right| = \lim_{n \rightarrow \infty} \frac{3}{n+1}$$

$$= 0 < 1$$

By Absolute Ratio Test, $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3^n}{n!}$ converges absolutely.

- **Example** Test for the convergence or divergence of $\sum_{n=1}^{\infty} (-1)^n \frac{n^n}{n!}$.

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} \frac{(n+1)^{\cancel{n+1}}}{(\cancel{n+1})!}}{(-1)^n \frac{n^n}{\cancel{n!}}} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1)^n}{n^n} \\
 &= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n \\
 &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n \\
 &= e > 1
 \end{aligned}$$

By Absolute Ratio Test, $\sum_{n=1}^{\infty} (-1)^n \frac{n^n}{n!}$ diverges.

Try Test for Divergence :

$$\lim_{n \rightarrow \infty} (-1)^n \frac{n^n}{n!} = ?$$

$$\text{Consider } \lim_{n \rightarrow \infty} \left| (-1)^n \frac{n^n}{n!} \right| = \lim_{n \rightarrow \infty} \frac{n^n}{n!}$$

$$= \lim_{n \rightarrow \infty} \frac{n \times n \times \dots \times n \times n \times n}{n \times (n-1) \times \dots \times 3 \times 2 \times 1}$$

$$= \infty$$

So, $\lim_{n \rightarrow \infty} (-1)^n \frac{n^n}{n!}$ does not exist

By Test for Divergence, $\sum_{n=1}^{\infty} (-1)^n \frac{n^n}{n!}$ diverges -

- **The Root Test**

Let $\sum a_n$ be a series of nonzero terms and suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$$

- (i) If $L < 1$, the series converges absolutely.
- (ii) If $L > 1$ (including $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \infty$), the series diverges.
- (iii) If $L = 1$, the test is inconclusive.

- **Example** Test the series $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$ for convergence or divergence.

$$\begin{aligned} \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{2n+3}{3n+2} \right)^n} &= \lim_{n \rightarrow \infty} \frac{2n+3}{3n+2} \times \frac{\frac{1}{n}}{\frac{1}{n}} \\ &= \lim_{n \rightarrow \infty} \frac{2 + \frac{3}{n}}{3 + \frac{2}{n}} \\ &= \frac{2}{3} < 1 \end{aligned}$$

$\therefore \sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$ converges absolutely by the Root Test.

By a **rearrangement** of $\sum a_n$, we mean a series obtained by simply changing the order of the terms. For example,

$$a_1 + a_2 + a_5 + a_4 + a_3 + a_{20} + a_6 + a_7 + a_{15} + \cdots$$

• Rearrangement Theorem

- If $\sum a_n$ is an absolutely convergent series with sum S , then any rearranged of $\sum a_n$ has the same sum S .
- If $\sum a_n$ is a conditionally convergent series and r is any real number, then there is a rearrangement of $\sum a_n$ that has a sum equal to r .

Consider the series :

$$1 - 1 + \frac{1}{2} - \frac{1}{2} + \frac{1}{3} - \frac{1}{3} + \frac{1}{4} - \frac{1}{4} + \cdots$$

$$\bullet \sum |a_n| = \sum \frac{2}{n} = 2 \sum \frac{1}{n} \quad \text{divergent due to harmonic series}$$

$$\bullet \sum a_n = 0 \quad \text{since} \quad \begin{aligned} S_1 &= 1 \\ S_2 &= 0 \\ S_3 &= \frac{1}{2} \\ S_4 &= 0 \\ S_5 &= \frac{1}{3} \\ S_6 &= 0 \\ &\vdots \end{aligned}$$

$$\lim_{n \rightarrow \infty} S_n = 0$$

Exercises Page 742 3, 5, 7, 9, 11, 13, 15, 25, 27, 31, 33, 35, 39, 43, 45.

$\therefore \sum a_n$ converges conditionally.

Consider the rearrangement :

$$1 + \left(\frac{1}{2} - 1\right) + \frac{1}{3} + \left(\frac{1}{4} - \frac{1}{2}\right) + \frac{1}{5} + \left(\frac{1}{6} - \frac{1}{3}\right) + \dots$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

$$= \ln 2 \quad \text{Ex 11.5 \#36 p. 737}$$

11.7 Strategy for Testing Series

1. $\sum \frac{1}{n^p}$ is a p -series. It converges if $p > 1$ and diverges if $p \leq 1$.
2. $\sum ar^{n-1}$ is a geometric series. It converges if $|r| < 1$, and diverges if $|r| \geq 1$.
3. If the series has a form that is similar to a p -series or a geometric series, then one of the comparison tests should be considered.
4. If you can see at a glance that $\lim_{n \rightarrow \infty} a_n \neq 0$, then the Test for Divergence should be used.
5. If the series is of the form $\sum (-1)^{n-1} b_n$, then the Alternating Series Test is an obvious possibility.
6. Series that involve factorials or other products (including a constant raised to the n^{th} power) are often conveniently tested using the Ratio Test.
7. If a_n is of the form $(b_n)^n$, then the Root Test may be useful.
8. If $a_n = f(n)$, where $\int_1^\infty f(x)dx$ is easily evaluated, then the Integral Test is effective.

- **Example** $\sum_{n=1}^{\infty} \frac{n-1}{2n+1}$.

$$\lim_{n \rightarrow \infty} \frac{n-1}{2n+1} = \lim_{n \rightarrow \infty} \frac{1 - \frac{1}{n}}{2 + \frac{1}{n}} = \frac{1}{2} \neq 0$$

Use Test for Divergence.

- **Example** $\sum_{n=1}^{\infty} \frac{\sqrt{n^3+1}}{3n^3+4n^2+2}$.

$$\text{Choose } b_n = \frac{\sqrt{n^3}}{3n^3} = \frac{1}{3n^{3/2}}$$

This is related to $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ a p-series.

Use comparison test.

- **Example** $\sum_{n=1}^{\infty} n e^{-n^2}$.

$f(x) = x e^{-x^2}$ continuous, positive, decreasing on $[1, \infty)$.

$\int_1^{\infty} x e^{-x^2} dx$ is easy to evaluate.

Use Integral Test.

- **Example** $\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{n^4 + 1}$.

$(-1)^n$ alternating

Use Alternating Series Test

- **Example** $\sum_{n=1}^{\infty} \frac{2^n}{n!}$

2^n : power n

$n!$: factorial

Use Absolute Ratio Test.

- **Example** $\sum_{n=1}^{\infty} \frac{1}{2 + 3^n}$

Choose $b_n = \frac{1}{3^n}$ which is related to $\sum_{n=1}^{\infty} \frac{1}{3^n}$ a

geometric series -

Use Comparison Test.

11.8 Power Series

- A **power series** in x is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots$$

- **Example** For what x does the power series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \dots$$

converge and what is its sum?

$$1 + x + x^2 + x^3 + x^4 + \dots$$

is a geometric series with

$$a = 1, \quad r = x.$$

The series converges when $|r| < 1$

$$\Rightarrow |x| < 1.$$

$$\text{The sum } \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \quad |x| < 1.$$

good to remember !

- **Example** For what values of x is the series

$$\sum_{n=0}^{\infty} \frac{x^n}{(n+1)2^n} = 1 + \frac{1}{2} \frac{x}{2} + \frac{1}{3} \frac{x^2}{2^2} + \frac{1}{4} \frac{x^3}{2^3} + \dots$$

convergent?

Use Absolute Ratio Test : series is convergent if

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{x^{n+1}}{(n+2)2^{n+1}}}{\frac{x^n}{(n+1)2^n}} \right| < 1$$

$$\Rightarrow \left(\lim_{n \rightarrow \infty} \frac{n+1}{n+2} \right) \frac{|x|}{2} < 1$$

$$\Rightarrow |x| < 2$$

Need to check for $|x| = 2$ separately as $\lim_{n \rightarrow \infty} \left| \frac{C_{n+1}}{C_n} \right| = 1$

for Ratio Test is inconclusive.

$$\underline{x=2} : \sum_{n=0}^{\infty} \frac{\cancel{2^n}}{(n+1)\cancel{2^n}} = \sum_{n=0}^{\infty} \frac{1}{n+1} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

diverges as it is a harmonic series.

$$\underline{x = -2} : \sum_{n=0}^{\infty} \frac{(-2)^n}{(n+1) 2^n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1}$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

converges as it is an alternating harmonic series.

$$\therefore \sum_{n=0}^{\infty} \frac{x^n}{(n+1) 2^n} \text{ converges when } -2 \leq x < 2.$$

$$[-2, 2)$$

- **Example** For what values of x is the series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ convergent?

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{x^{n+1}}{(n+1)!}}{\frac{x^n}{n!}} \right| < 1$$

0
||

$$\Rightarrow \left(\lim_{n \rightarrow \infty} \frac{1}{n+1} \right) |x| < 1$$

$$\Rightarrow x \in \mathbb{R}$$

$$\therefore \sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ converges everywhere.}$$

- **Example** For what values of x is the series $\sum_{n=0}^{\infty} n! x^n$ convergent?

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)! x^{n+1}}{n! x^n} \right| < 1$$

∞
||

$$\Rightarrow \left(\lim_{n \rightarrow \infty} (n+1) \right) |x| < 1$$

$$\Rightarrow x = 0$$

$$\therefore \sum_{n=0}^{\infty} n! x^n \text{ converges when } x = 0.$$

- **Theorem**

The set on which a power series $\sum_{n=0}^{\infty} c_n x^n$ converges is always an interval of one of the following three types:

- (i) The single point $x = 0$.
- (ii) An interval $(-R, R)$, plus possibly one or both end points.
- (iii) The whole real line.

- The number R in case (ii) is called the **radius of convergence** of the power series.
- Radius of convergence in each case:
 - (i) $R = 0$
 - (ii) $R = R$
 - (iii) $R = \infty$
- The **interval of convergence** of a power series is the interval that consists of all values of x for which the series converges.
- Interval of convergence in each case:
 - (i) $\{0\}$
 - (ii) $(-R, R)$ or $[-R, R)$ or $(-R, R]$ or $[-R, R]$
 - (iii) $(-\infty, \infty)$

More generally, a series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

is called a **power series in $(x-a)$** or a **power series centered at a** .

- **Theorem**

The set on which a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ converges is always an interval of one of the following three types:

- (i) The single point $x = a$.
- (ii) An interval $(a-R, a+R)$, plus possibly one or both end points.
- (iii) The whole real line.

- Radius of convergence in each case:

- (i) $R = 0$
- (ii) $R = R$
- (iii) $R = \infty$

- Interval of convergence in each case:

- (i) $\{a\}$
- (ii) $(a-R, a+R)$ or $[a-R, a+R)$ or $(a-R, a+R]$ or $[a-R, a+R]$
- (iii) $(-\infty, \infty)$

- Note that if $\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| = L$, then $R = \frac{1}{L}$.

- **Example** Find the interval of convergence and radius of convergence of

$$\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}.$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(-3)^{n+1} x^{n+1}}{\sqrt{n+2}}}{\frac{(-3)^n x^n}{\sqrt{n+1}}} \right| < 1$$

$$\Rightarrow \left(\lim_{n \rightarrow \infty} \frac{\sqrt{n+1}}{\sqrt{n+2}} \right) 3 |x| < 1$$

$$\Rightarrow |x| < \frac{1}{3}$$

$$\underline{x = \frac{1}{3}} : \quad \sum_{n=0}^{\infty} \frac{(-3)^n \left(\frac{1}{3}\right)^n}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$$

$$\text{Let } b_n = \frac{1}{\sqrt{n+1}}$$

$$\cdot \quad \frac{1}{\sqrt{n+1}} > \frac{1}{\sqrt{n+2}} \quad \text{for } n \geq 1$$

$$\cdot \quad \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1}} = 0$$

By Alternating Series Test, $\sum_{n=0}^{\infty} \frac{(-3)^n \left(\frac{1}{3}\right)^n}{\sqrt{n+1}}$ converges.

$$\underline{x = -\frac{1}{3}} : \quad \sum_{n=0}^{\infty} \frac{(-3)^n \left(-\frac{1}{3}\right)^n}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n+1}} = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

This series diverges as it is a p -series with
 $p = \frac{1}{2} \not> 1$.

$$\therefore \text{Interval of convergence} = \left(-\frac{1}{3}, \frac{1}{3}\right]$$

$$\text{Radius of convergence} = \frac{1}{3}$$

- **Example** Find the interval of convergence and radius of convergence of

$$\sum_{n=0}^{\infty} \frac{n(x+2)^n}{3^{n+1}}.$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(n+1)(x+2)^{n+1}}{3^{n+2}}}{\frac{n(x+2)^n}{3^{n+1}}} \right| < 1$$

$$\Rightarrow \left(\lim_{n \rightarrow \infty} \frac{n+1}{n} \right) \frac{1}{3} |x+2| < 1$$

$$\Rightarrow |x+2| < 3$$

$$\underline{x+2=3} : \sum_{n=0}^{\infty} \frac{n 3^n}{3^{n+1}} = \sum_{n=0}^{\infty} \frac{n}{3}$$

$$\lim_{n \rightarrow \infty} \frac{n}{3} = \infty \neq 0, \text{ the series diverges by}$$

Test for Divergence.

$$\underline{x+2=-3} : \sum_{n=0}^{\infty} \frac{n(-3)^n}{3^{n+1}} = \sum_{n=0}^{\infty} (-1)^n \frac{n}{3}$$

Since $\lim_{n \rightarrow \infty} (-1)^n \frac{n}{3}$ does not exist,

by the Test for Divergence, the series diverges.

Interval of Convergence : $(-5, 1)$

Radius of Convergence : 3